no.9 ETL (Extract, Tranform, load).

ETL - it is trans-ferring process of data to data-werehouse

Extraction - it is the first phase where all ~~collet~~ collected data from regular database (raw data) is collected e.g files, emails or API's. Extraction is meant to retrieve data without altering its format for further processing.

Tranformation - this is the second phase where raw data collected/extracted is cleaned, standarsized and ~~enhance~~ enhanced. This is where duplicates are cleared, data is summarised and validated.

Loading - final phase/stage where extracted data, cleaned summerised data is ~~depo~~ deposited into data warehouse, data lake. Loading has 3x methods known as;

- Initial load
- Incremental load
- full refresh

No. 10 Common data storage used in analytics

a) CSV (comma-separated values) it is a plain text file configuration that ~~used~~ uses commas to separate values.

b) Json ?? (I struggled to understand defination and therefore could not find right words to describe it)

Parquet - it is widely adopted columner format designed for big analytics, offering Compression, which enables teams freedom to balance CPU (central Processing).

No. 11 AI (artificial Intelligence)

AI - refers to to computer system capable of carrying out complex tasks that habitually only humans could do.

Narrow (weak) AI	General (Strong) AI
specialised in performing a specific task or set of tasks	Process human like intelligence
Limited to what is set or programmed for and lacks understanding beyond its specific domain	Capable of learning, reasoning and perform any intellectual task.

Narrow AI Examples	General AI examples
Recommendation algorithms (used by META to track likes and searches of an ~~at~~ instagram and facebook users - image recognition Systems (like facial recognition - technolowledge tha exists on to unlock phone or banking app	it is able to handle different tasks like (reasoning about the planet, natural language, adapting different situations same way as humans.

~~Note~~ MACHINE LEARNING (ML)

It is an artificial intelligence machine which enables scalability (expansion, extension, adaptation, flexibility, upgrade etc.) without manual intervention ~~are~~ Once trained/programed, ML systems can scale their operations without extra effort

Difference between Traditional programming and Machine Learning

Traditiona Programming	Machine Learning
• Relies on explicity defined rules and logic written by developers	• It is data driven approach where algorithms ~~lean~~ learn patterns from data to make ~~decio~~ decisions
• Suitable for problem solving with clear, deter-ministic logic	• Better for ~~deal~~ dealing ~~of~~ with complex problems where patterns and relationships are not ~~e~~ evident. Such as image-recognition.
• Outcome is highly predictable if inputs and logic are ~~known~~ known.	Compared to traditional learning (ML) is an automated process
$\boxed{\text{input}} + \boxed{\text{Program}} = \boxed{\text{Output}}$	$\boxed{\text{input}} + \boxed{\text{OUTPUT}} = \boxed{\text{Program}}$

~~Reinforcement Learning~~

Three types of machine learning

1) Supervised Machine Learning

when a model is trained on labled "Dataset". Labled datasets consists of input and output parameters. Supervised learning has one of the categories called,

a) Classification

↳ this category predicts categorical outputs, it assigns data into predefined classes like Spam/non spam emails.

b) Regression - on this category supervised machine learning predicts continuous values, such house prices or product sales.

2) Unsupervised Machine Learning

It works with unlabled data, Algorithm finds hidden patterns, groups or relation within the data on its own; does not have any data training set. It processes/predicts patterns by;

a) Clustering - grouping points clusters based on similarity. This technique method is functional for identifying patterns and ~~relationsphis~~ relationships with no need/~~he~~ help for labled examples

b) Dimensionality Reduction Techniques - assist to reduce number of features while preserving important information.

c) Reinforced Learning - trains an agent to make sequence of decision through trial and error. this machine ~~tea~~ learning, learns the best action for each state on expected rewards, methods is called Q-Learning.

Types of reinforcement Learning

- Positive Reinforcement – rewards behaviour (i.e gives points for correct answer.)
- Negative Reinforcement – removes negative outcome to encourage good action.

No. 3 Deep Learning

it is transforming the way machines understand ~~lea~~ learn and interact with complex data. Deep learning impersonates neural networks of human brains. It enables computers to autonomously uncover patterns and make informed decisions from vast amounts of unstructured data

Machine Learning ←	"Relation"	→ Deep Learning
• Take less time to train.	←→	~~takes~~ takes more time to train
• Less complicated	←→	Highly complicated
• Easier to understand and explain.	←→	Hard to interpret
• Spam ~~dect~~ detection recommendation System	←→	image recognition, NLP, speech recognition

Neural Networks

it is a machine learning algorithm designed to recognise patterns and make decision without programming rules. They work in layers namely;

a) Hidden layer – this layers perfom most performs/processes heavy lifting. The work hidden ~~layers~~ layer is to transform each that consists of neurons into something that the output of model (Results of the model)

b) output layer - this is the last/final layer which produces the output of the model. (Results of the model.)

No. 14 Natural Language Processing

It is a field of intelligence (AI) that enables computers to understand, interprete and respond to ~~be~~ human language. It is a way of teaching computer to mimic human language or a way of communicating.

Three real world applications of NLP - I use in my daily

a.) Generating - when I compose an email as I start typing there will be some texts predictions for me which aligns to email context I am writing.

b) Translation - Now I know that when I translate sesotho words to another language it is with help of NLP.

c) CHATGPT META AI - what is mostly used lately to do researches etc. spelling checker and grammar corrector.